

Yash Thube

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Experience

Future Sportler – Remote	01/2026 – Present
<i>AI/ML Lead</i>	07/2026 – Present
<i>ML Research Engineer</i>	01/2026 – 07/2026
• Building end-to-end zero-shot markerless biomechanics pipelines—spanning pose tracking, action recognition, 3D reconstruction, and VLM-driven feedback from single-view consumer video. • Designed occlusion-robust identity tracking and multi-object fusion with custom geometric and covariance-weighted evidence gating for scenes where bodies or objects overlap heavily, where standard trackers collapse multiple entities into a single detection or lose identity through contact. • Built anatomically-constrained inverse kinematics and 3D human mesh recovery so recovered joint angles respect real physical limits instead of coming straight off noisy 2D keypoints, alongside classical camera calibration (homography estimation, joint bundle adjustment) to convert pixel motion into real-world metric units without a dedicated rig. • Built physics-grounded event detection and state estimation — finite-state machines, kinematic-chain sequencing, hysteresis-based debouncing, and rules-constrained inference for recovering physically valid state from partial evidence — plus optimization-based strategy planning (CEM) against a physics simulator. • Fine-tuned, quantized and merged VLM adapters for grounded, session-relative critique, paired with diffusion-based correction overlays composited back onto source video. Cross-processd GPU semaphores via flock, batched Blender + RIFE frame interpolation rendering to amortize shader compilation cost.	
Machine Learning Engineer , Hudl India – Pune, MH	03/2024 – 12/2024
• Built a multi-camera YOLO-based pipeline for real-time player detection across broadcast angles, with temporal tracking to handle occlusion. • Fine-tuned VLMs and video transformers on custom sports data for semantic action recognition. • Scaled DeepSORT-based MOT to 92% precision in crowded scenes; mapped 2D detections to 3D space for kinematic localization. • Engineered inference optimizations via model quantization and ONNX runtime deployment, cutting latency and maximizing FPS for real-time, multi-camera action tracking.	
Deep Learning Collaborator , Omdena – Remote	10/2023 – 02/2024
• Built BERT/RobERTa multimodal classifiers and applied RLHF (PPO) for clinical safety alignment. • Trained ResNet-based CNN pipelines on mel-spectrograms for real-time vocal stress detection.	
Computer Vision Collaborator , AI Accelerator Institute – Remote	02/2023 – 06/2023
• Worked on segmentation and SSL algos for vision (SimCLR, MoCo-v2, BYOL, DINO).	

Research & Applied Interests

Broadly — 3D Vision & Embodied AI (scene understanding, multi-view geometry, 3D/4D reconstruction, depth/pose estimation), representation learning (SSL, 3DGS, NeRFs, point clouds), and model-based RL / world models (predictive & generative modeling, spatial-causal reasoning).

Stack used — PyTorch, OpenCV, NumPy, Hugging Face (Transformers, Diffusers, PEFT, TRL), OpenMMLab (MMPose, MMAction2), PyTorch3D, Open3D, gsplat, Trimesh, SMPL-X, GVHMR, OpenSim, Panda3D, SAM2 / SAMURAI, CoTracker3, TrackNetV3, YOLO, BoT-SORT, ByteTrack, OC-SORT, DeepSORT, Blender (BPY), FFmpeg, ONNX, Docker, Linux GPU clusters (CUDA, SSH) MuJoCo, MuZero, EfficientZero, TransformerLens.

Projects & Research

- **MATS (arXiv preprint)** – A behavioral audit toolkit to detect pathological truth bias in VLMs, experiments include activation patching to causally localize failures in cross-attention layers and pooled representations across LLaVA, CLIP, and Qwen-VL architectures.
- **LC-JEPA** – Self-supervised JEPA-style pretraining on ZTF light curves for changing-look AGN discovery; controlled 5-seed objective ablation (reconstruction 0.888 AUC vs. latent-prediction 0.870, brightness-confound-controlled) with 12× top-20 enrichment over base rate on 561 confirmed CLAGN.
- **MBRL** – Single-file implementations of model-based reinforcement learning algorithms across different environments, from discrete control to pixel-based DMControl tasks.
- **Multimodal/VLMs Research Hub** – Curated technical resource on VLMs and multimodal learning — papers, datasets, benchmarks, open problems.

Github

Education

Savitribai Phule Pune University (SPPU)

2025

B.E. Computer Engineering